

Displacement and velocity



1

a $v(t) = 12t$

b

i Yes ii No iii No iv Yes

c 48 m/s

2

a $v(t) = 9t^2 - 8t$

b 20 m/s

3

a $v(t) = 6t + 5$

b 29 m/s

4

a $v(t) = \frac{9}{\sqrt{t}}$

b 3 m/s

5

a 67.5 m/s

b $v(t) = \frac{3t^2}{2} + 6t$

c 67.5 m/s

d $t = 2$ s

6

a 0 m/s

b $(-9.8x + 11.27)$ m/s

c 11.27 m/s

d 5.635 m/s

e 0 m/s

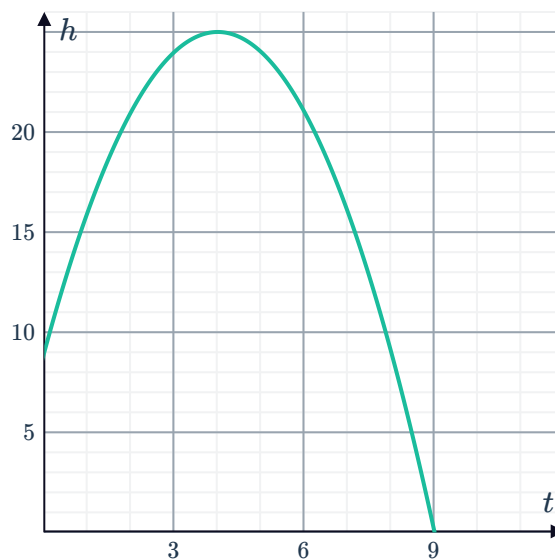
7

a 9 m

b $h = 25$ m

c $t = 9$ s

d





8

a $h = 16 \text{ m}$

b $t = 6 \text{ s}$

c $\frac{dh}{dt} = -32 \text{ m/s}$

9

a -4 m

b $v(t) = 6t^2 - 14t + 3$

c 3 m/s

d 1 m/s

10

a 9 m

b $v(t) = 32t + 24$

c 24 m/s

d 24 m/s

11

a 0 m

b $t = 4 \text{ s}$

c 40 m

12

a 0 m

b $v(t) = -4t - 20$

c -20 m/s

d 20 m/s

13

a $\frac{ds}{dt} = 12 - 9.4t$

b $\frac{d^2s}{dt^2} = -9.4 \text{ m/s}^2$

14

a $\frac{dh}{dt} = -13 \text{ m/s}$

b $\frac{d^2h}{dt^2} = -10 \text{ m/s}^2$

15 $a = 1, b = 8$

Acceleration

16

a $v(t) = 4t - 4$

b $t = 1 \text{ s}$

c $a(t) = 4$

17

a $v(t) = 2t - 7$

b -7 m/s

c $a(t) = 2$

d 2 m/s^2

e $t = 5 \text{ s}$

f $t = 5, 2 \text{ s}$

18

a $\frac{d^2s}{dt^2} = t + \frac{1}{2}$

b $\frac{1}{2} \text{ m/s}^2$

c $\frac{33}{2} \text{ m/s}^2$

19

a $\frac{d^2s}{dt^2} = 12t^2 + 2$

b 2 m/s^2



c 110 m/s^2

20

a $a = 38 - 12t$

b -10 m/s^2

c The particle is slowing down.

21

a $t = 4 \text{ s}$

b 0 m/s

c $t = 2 \text{ s}$

d 12 m/s

e The velocity is constant. The particle is neither speeding up or slowing down.

22

a $t = \frac{3}{2} \text{ m/s}$

b $a(2) = 20 \text{ m/s}^2$

c The particle starts moving to the right, but eventually comes to rest. After instantaneously coming to rest, it then continues to move to the right.

23 $v = -\frac{25}{2} \text{ m/s}$

24 $x = -12 \text{ m}$

25

a $t = 8 \text{ s}, 3 \text{ s}$

b $a(t) = 2t - 11$

c -7 m/s^2

26

a $t = \frac{8}{5} \text{ s}, 3 \text{ s}$

b $a(t) = 10t - 23$

c 7 m/s^2

27

a $t = 2$

b $x''(t) = -6t - 2$

c $x''(2) = -14 \text{ m/s}^2$

d 4 m/s

28

a $d' = 1.72t - 0.0252t^2$

b $t = 68 \text{ s}$

c 1335 m

d $d' = 29 \text{ m/s}$



29

a $\frac{9}{10} \text{ s}$

b $s'(t) = u + at$

c $t = -\frac{u}{a}$

d $a = -\frac{50}{9} \text{ m/s}^2$

Graphs of displacement, velocity and acceleration

30

a 0 m

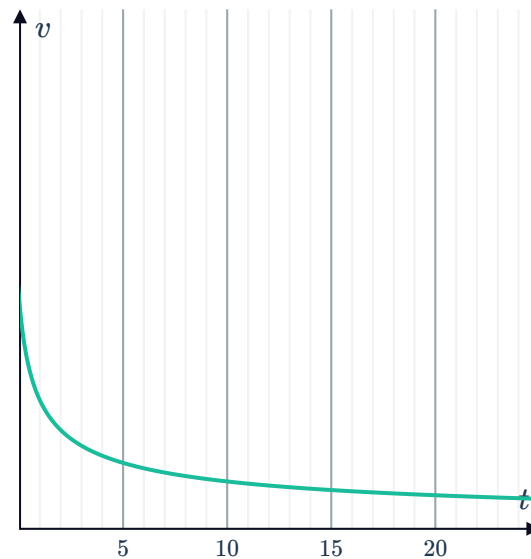
b 30 m

31

a -6 m

b 6 m

32



33

a 3, 5, 10

b $0 < t < 4, t > 8$

c 4, 8

34

a 6, 11

b $6 < t < 11$

35

a i -2 m ii 8 m

b -3 m/s

c 2 m/s

36

a 2s

b $0 < t < 2$

c 4 m

d $x = -t(t - 4)$

e 4 s

f $\frac{dx}{dt} = -2t + 4$

g -2 m/s

37

a 9 m

b $6 \text{ s}, 15 \text{ s}$



c To the left

d 11 s

e $-\frac{8}{5} \text{ m/s}$

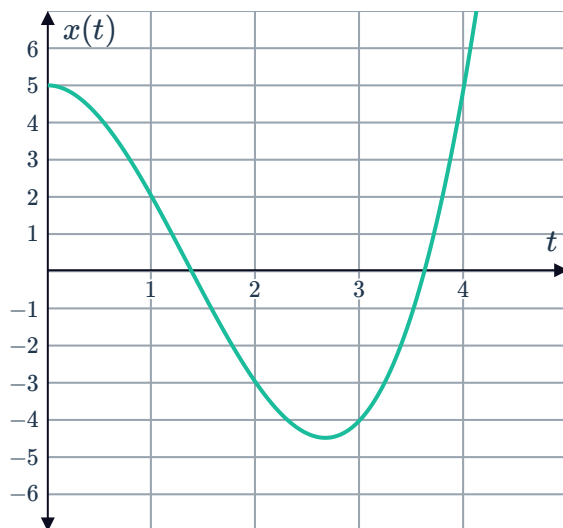
38

a $v(t) = 3t^2 - 8t$

b $t = 0 \text{ s}, \frac{8}{3} \text{ s}$

d $\frac{256}{27} \text{ units}$

c



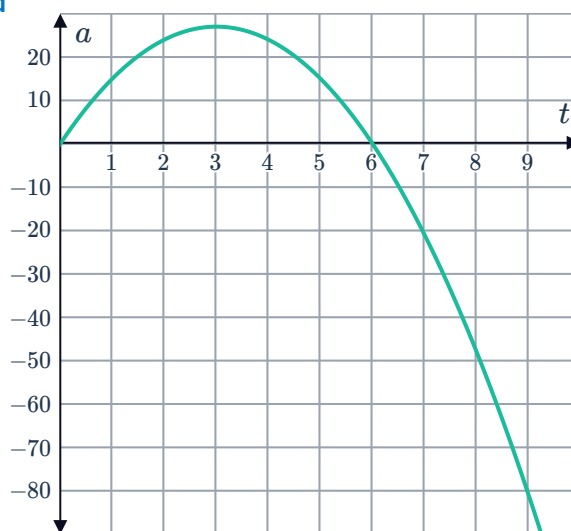
39

a $a = 18t - 3t^2$

c 108

b $t = 6 \text{ s}$

d



e 81

40

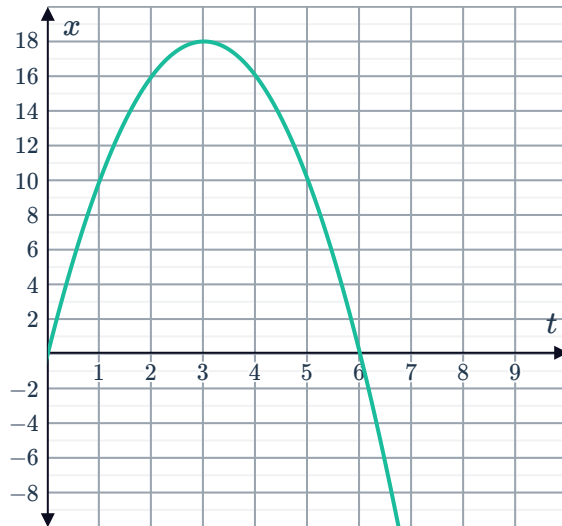
a $v = 12 - 4t$



b $t = 3$ s

c

d -54 m



e 90 m

41

a $x'(t) = 3t^2 - 18t + 24$

b $x''(t) = 6t - 18$

c

i

+	$t < 2$
0	$t = 2$
-	$2 < t < 4$
0	$t = 4$
+	$t > 4$

ii

-	$t < 3$
0	$t = 3$
+	$t > 3$

d 2 s, 4 s

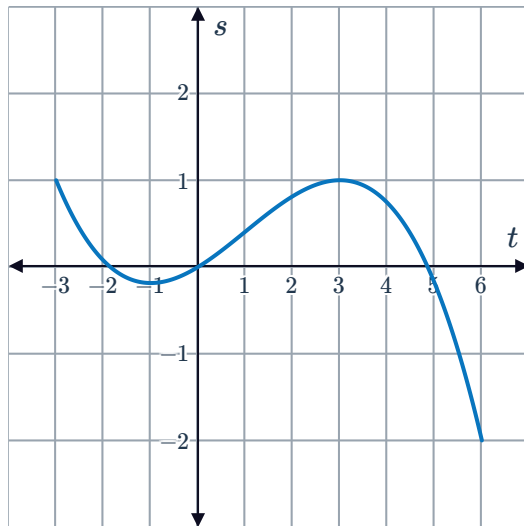
e $t < 2, 3 < t < 4$

f $t < 3$

g 28 m

42

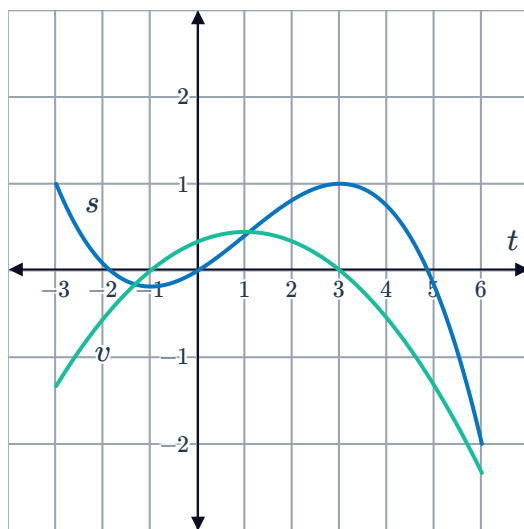
a



b

$$\frac{ds}{dt} = \frac{1}{3} + \frac{2}{9}t - \frac{1}{9}t^2$$

c



d 4 s

e $-\frac{5}{27} \text{ m}$

f 1 m

g $\frac{8}{27} \text{ m/s}$

Displacement and acceleration from velocity

43

a

i $0 \leq t \leq 10$

ii No values of t .

b

i No values of t .

ii $0 \leq t \leq 10$

c

i $0 \leq t \leq 4, 8 \leq t \leq 10$

ii $4 \leq t \leq 8$

d

i $7 < t < 10$

ii $0 < t < 5$



44

a

i 11 units

ii 19 units

b

i 6 units

ii 18 units

c

i -3 units (or 3 units to the left)

ii 15 units

d

i 1.75 units

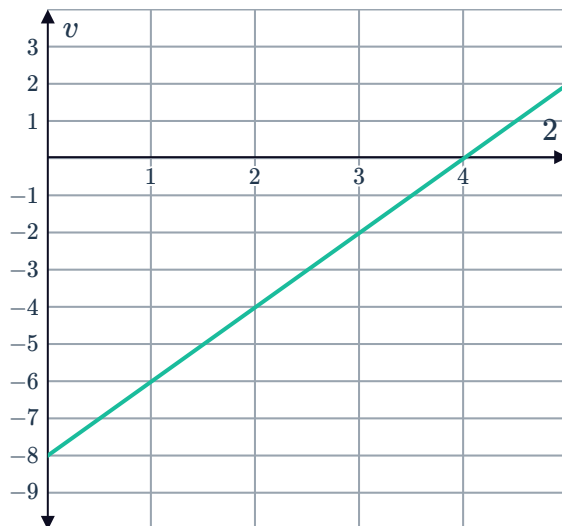
ii 19.75 units

45 Example answers:

- Because both are equal to the area of the triangle above the horizontal axis from $t = 0$ to $t = 6$.
- Because the particle only moved in one direction during the first six seconds.

46

a



b -15 units or 15 units to the left.

c 17 units.

47

a

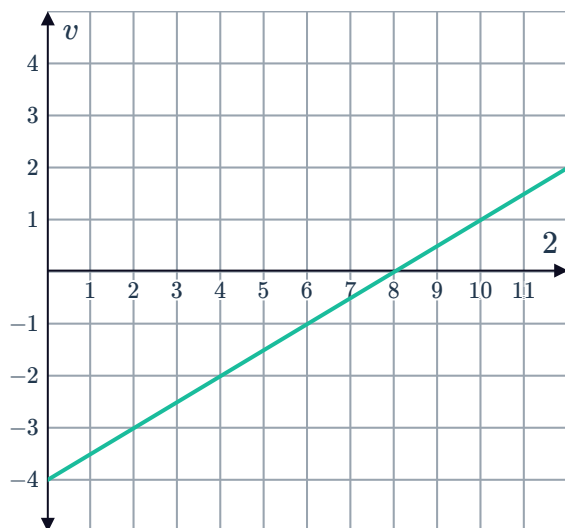


b -10 units or 10 units to the left.

c 26 units.

48

a



b -3 units or 3 units to the left.

c 5 units.